Abstract Algebra

SF1078

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0.0.1. Cosets

Definition 0.1 (Coset)

Let $H \subseteq G$ be a subgroup of a group G. For any $g \in G$, the *left coset* of H in G is

$$gH = \{g * h \mid h \in H\}.$$

Similarly, the *right coset* is

$$Hg = \{h * g \mid h \in H\}.$$

Theorem 0.1 (Lagrange's)

Let G be a finite group and $H \subseteq G$ a subgroup. Then the order of H divides the order of G.

Corollary

If G is of prime order, it has no nontrivial subgroups.

Theorem 0.2 (Lagrange's, Generalised)

Regardless of whether G is finite or infinite,

$$|G| = |G : H||H|.$$

0.0.2. Symmetry

Definition 0.2 (Isometry)

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is an *isometry* if it preserves distances, that is,

$$d_X(x_1, x_2) = d_Y(f(x_1), f(x_2)) \quad \forall x_1, x_2 \in X.$$

Definition 0.3 (Symmetry)

A *symmetry* of an object $\mathcal{O} \subseteq X$ is a metric space (X, d) an isometry $f : X \to X$ with the image $f(\mathcal{O}) = \mathcal{O}$.

Informally, a symmetry of a geometric object is a rearrangement of the object preserving the arrangement of its sides and vertices as well as its distances and angles.

Examples. Some types of symmetries:

(i) **Reflectional** Reflection across a hyperplane (a line in $\mathbb{R}^2$ or a plane in $\mathbb{R}^3$).
(ii) **Rotational** Rotation within a plane ($\mathbb{R}^2$ is rotated around a fixed point, while $\mathbb{R}^3$ is rotated around an axis).
(iii) **Translational** All points are shifted by a constant vector.
(iv) **Identity** No change.

Definition 0.4 (Rigid Motion)

A *rigid motion* is a map $f : \mathbb{R}^n \to \mathbb{R}^n$ such that

$$\|\mathbf{u} - \mathbf{v}\| = \|f(\mathbf{u}) - f(\mathbf{v})\| \quad \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n.$$

Equivalently, it is a surjective isometry between Euclidean spaces

Example. A symmetry of an object in an Euclidean space $X = \mathbb{R}^n$ is also a rigid motion.

Definition 0.5 (Dihedral Group)

The *dihedral group* R_n is the group of symmetries (or equivalently rigid motions) of an n-gon. Its degree is n and its order is $2n$.

1. Sets

1.0.1. Set theory

Definition 1.0.1 (Tuple)

A tuple (or sometimes list) is a finite, ordered collection of elements, possibly with repetitions.

An n-tuple is of length n. A 2-tuple is called an *ordered pair*.

1.0.2. Morphisms

Definition 1.0.2 (Map)

Subsets of the Cartesian product $A \times B$ are called *relations*. A map is a *well-defined* relation written

$$f \subseteq A \times B \quad \text{or} \quad f : A \to B \quad \text{or} \quad A \xrightarrow{f} B$$

and is

- *surjective* or *onto* if $f(A) = B$, which is equivalent to being *right invertible* with $f \circ f^{-1} = \mathrm{id}_B$.
- *injective* or *one-to-one* if $a_1 \neq a_2 \implies f(a_1) \neq f(a_2)$, which is equivalent to being *left invertible* with $f^{-1} \circ f = \mathrm{id}_A$.
- *bijective* or a *one-to-one correspondence* if both above apply, which is equivalent to being *invertible*.

Definition 1.0.3 (Endomorphism)

Let X be some mathematical object. A map $\varphi : X \to X$ is an *endomorphism*. Should φ simultaneously be bijective it is moreover an *automorphism* and, if X is a set, a *permutation*. The most trivial automorphism is the (the *identity* map id_X with

$$\mathrm{id}_X(x) = x \qquad \forall x \in X.$$

Definition 1.0.4 (Homomorphism)

A *homomorphism* is a map between algebraic structures that preserves the structure.

For example, a group homomorphism $f : G \to H$ satisfies

$$f(a * b) = f(a) * f(b)$$

for all $a, b \in G$.

Definition 1.0.5 (Isomorphism)

An *isomorphism* is a bijective homomorphism.

If there exists an isomorphism between two algebraic structures, they are said to be *isomorphic*.

1.0.3. Equivalence

Definition 1.0.6 (Equivalence Relation)

An *equivalence relation* on a set A is a relation $R \subseteq A \times A$ with the following properties:

(i) **reflexive** $(a, a) \in R \qquad \forall a \in A$
(ii) **symmetric** $(a, b) \in R \implies (b, a) \in R$
(iii) **transitive** $(a, b), (b, c) \in R \implies (a, c) \in R$

If $(x, y) \in R$ we write $x \sim y$.

Definition 1.0.7 (Partition)

A *partition* $\mathcal{P}$ of a set A is a collection of non-empty, disjoint subsets of A whose union is A.

Definition 1.0.8 (Equivalence Class)

Let R be an equivalence relation on a set A. The *equivalence class* of $a \in A$ is

$$[a] = \{b \in A \mid b \sim a\}.$$

The set of all equivalence classes is written

$$A \nearrow R$$

and is called the *quotient set*.
The equivalence classes form a partition of A.

Definition 1.0.9 (Quotient Map)

Let R be an equivalence relation on A. The map $\pi : A \to A \nearrow R$ defined by

$$\pi(a) = [a]$$

is called the *quotient map*.
It is always surjective.

2. Groups

2.0.1. Definition

Definition 2.0.1 (Group)

A *group* $(G, *)$ is a nonempty set G with a binary operation

$$* : G \times G \to G$$

such that for all $a, b, c \in G$, it satisfies

(i) (Closure) $a * b \in G$
(ii) (Associativity) $(a * b) * c = a * (b * c)$
(iii) (Identity) $\exists e \in G : e * a = a * e = a$
(iv) (Inverse) $\exists a^{-1} \in G : a * a^{-1} = a^{-1} * a = e$

Additionally, a group is said to be abelian or *commutative* if it is
(v) (Commutative) $a * b = b * a$

Tip

The proof of (iii) and (iv) can be shortened if (v) is proven.

Note

Defying conventional wisdom, *abelian* is usually not capitalized.

2.0.2. Subgroups

Let $(G, *)$ be a group.

Definition 2.0.2 (Subgroup)

A subset $H \subseteq G$ is a *subgroup* if it is itself a group under the operation inherited from G.

Tip

It follows that H is a subgroup if
(i) $H \neq \emptyset$
(ii) $a * b^{-1} \in H \quad \forall a, b \in H$

Definition 2.0.3 (Cyclic Subgroup)

For all $a \in G$, the set

$$\langle a \rangle = \{a^k \mid k \in \mathbb{Z}\}$$

with the operation $*$ is a *cyclical subgroup* of $(G, *)$.

Definition 2.0.4 (Cyclical Group)

We say $(G, *)$ is a *cyclical group* if $\exists a \in G : \langle a \rangle = G$.

2.1. Permutations

2.1.1. Transpositions

Definition 2.1.1 (Transposition)

A transposition of the elements $i, j \in \{1, ..., n\}$ is the permutation π_{ij} with the cycle notation (ij).

Theorem 2.1.1

A permutation $\sigma \in S_n$ can be written as a composition of transpositions.

Theorem 2.1.2

The identity $\mathrm{id} \subset S_n$ may only be written as an even number of transpositions.

Definition 2.1.2 (Sign of a Permutation)

Let n be the number of transpositions in any decomposition of a permutation $\sigma \in S_n$. The sign of σ is $(-1)^k$. Hence,

- if k is even then $\mathrm{sgn}(\sigma) = +1$ and σ is said to be even.
- if k is odd then $\mathrm{sgn}(\sigma) = -1$ and σ is said to be odd.

Definition 2.1.3 (Fixed Point)

A fixed point is invariant under a transformation such as a permutation.

2.1.2. Symmetric Groups

Definition 2.1.4 (Symmetric Group)

The *symmetric group* of the finite set $X = \{1, 2, ..., n\}$ is

$$S_n = \{\sigma : X \to X \mid \sigma \text{ is bijective}\}$$

and consists of $n!$ permutations, which form a group under the operation of composition called the *symmetric group on* n *letters*.

Definition 2.1.5 (Permutation Group)

A subgroup of a symmetric group S_n is a *permutation group*.

Definition 2.1.6 (Alternating Group)

The *alternating group*

$$A_n = \{\sigma \in S_n \mid \mathrm{sgn}(\sigma) = 1\}$$

is the subgroup of S_n consisting of all even permutations.

It has $n!/2$ elements for $n \geq 2$.

Theorem 2.1.3

The sign map $\mathrm{sgn} : S_n \to \{1, -1\}$ is a group homomorphism.
The kernel of this homomorphism is the alternating group

$$\ker(\mathrm{sgn}) = A_n.$$

3. Fields

Definition 3.1 (Field)

A *field* is a set F with two binary operations $+$ and $\cdot$ such that
(i) $(F, +)$ is an abelian group with identity 0.
(ii) $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1.
(iii) Multiplication distributes over addition:

$$\text{for all } a, b, c \in F, \; a \cdot (b + c) = a \cdot b + a \cdot c.$$

Examples. $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{R}(x), \mathbb{Q}(i), \mathbb{F}_p$ if p is prime.

Definition 3.2 (Vector Space)

A vector space over a field F is a set V with two operations:

- Vector addition $+ : V \times V \to V$
- Scalar multiplication $\cdot : F \times V \to V$

These must satisfy

(i) (Closure under addition) $\mathbf{u} + \mathbf{v} \in V$
(ii) (Distributivity over vectors) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
(iii) (Associativity) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
(iv) (Identity) $\mathbf{u} + \mathbf{0} = \mathbf{u}$

(v) (Inverse) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
(vi) (Commutativity) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

(vii) (Closure under scalar multiplication) $a\mathbf{u} \in V$
(viii) (Distributivity over scalars) $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$
(ix) (Associativity) $a(b\mathbf{u}) = (ab)\mathbf{u}$
(x) (Identity) $1\mathbf{u} = \mathbf{u}$

Note that requiring closure is often omitted.